



A real time cloud-based framework for glaucoma screening using EfficientNet

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Abstract

These days one of the major causes of partial or complete blindness that has affected a majority of people all around the world is glaucoma. Glaucoma is caused as a result of increased fluid pressure inside the optic nerves called intra ocular pressure. A real time cloud-based framework for screening the glaucoma suspect's retinal fundus images as received by the people on the public cloud, is proposed in this paper. In the proposed framework the existence of glaucoma and analysis of the retinal fundus images is achieved by deep learning technique and convolutional neural network respectively. EfficientNet and UNet++ models are used to identify the presence of glaucoma. On comparing our framework to various state-of-the-art models and quantitative assessment are performing on various benchmark datasets like RIM-ONE and DRISHTI-GS1, it was found that the proposed framework is scalable, location independent, and easily accessible to one and all due to the cloud platform.

Keywords Cloud-based · Glaucoma · Convolutional neural network · Real-time · EfficientNet

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1 Introduction

The eye is the most important organ that allows the vision. The eye contains the fluid material known as aqueous and vitreous humor that maintains fluid pressure inside the eye called intraocular pressure. Glaucoma is as a result of intra ocular pressure that affects the optics nerves resulting partial or complete vision loss. The visual signals are propagated from eye to the brain via optic nerves [45]. Glaucoma is detected using most common factor optic Cup-to-Disc Ratio (CDR) by identifying optic cup and disc from retinal fundus images. The CDR value decides the presence of glaucoma in the eye. Fundus cameras are used captured eye images that is used for separating various features like optics cup, disc, fovea and macula as displayed in Fig. 1.

World Health Organization (WHO) reports that 79 million peoples are infected by glaucoma in 2020 and this number increases to 112 million by 2040 [31]. The manual inspection by some expert is time taking, expensive process and beyond the reach of every person. Therefore, cloud based real time glaucoma identification system is required that automatically detect the presence of glaucoma in the eye [30]. Figure 2 shows the healthy and infected eye. CDR value is higher for glaucoma infected eye. Feature detection and classification is expensive process in digital image processing and it effects can be reduced using deep learning (DL) models which train their model on trained labeled images and classify the unknown image based on training models. Number of DL models like VGGNet [35], ResNet [18], GoogleNet [43] has been implemented for the same purpose.

Fig. 1 Different eye features used in glaucoma screening [29]

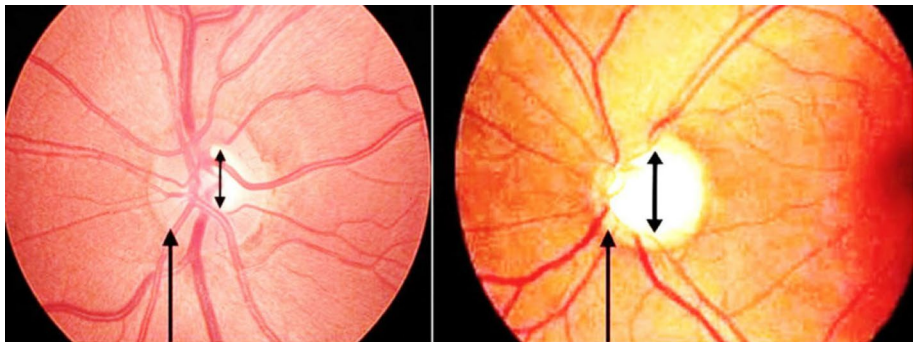
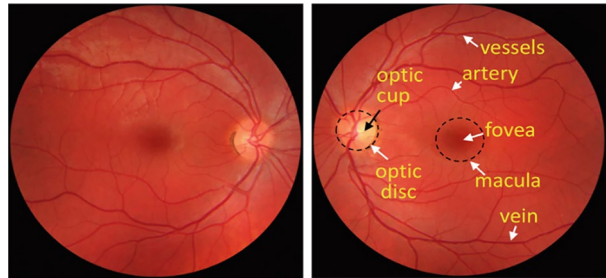


Fig. 2 Left non-infected and right glaucoma infected fundus image from DRISHTI-GS dataset

The proposed paper recommends a real-time cloud-based screening system to detect the presence of glaucoma using the EfficientNet [44] model. Although various state-of-the-art glaucoma detection techniques are giving quite good results. To fulfil the need of the large population, the conventional system is not applicable. To reach to the masses, we propose a cloud-based real-time screening system for retinal fundus images for identifying the presence of glaucoma in the eye. The recommended framework contains preprocessing of retinal fundus image and classify whether healthy or glaucoma infected eyes using the DL (EfficientNet) model. The proposed system is useful for general healthcare center where there is a lack of trained ophthalmologists.

1.1 Problem statement

- A real time cloud-based framework for glaucoma screening system.
- For segmentation of optic cup and optic disc utilized EfficientNet [44] as encoder and UNet++ [52] as decoder for classification.
- Further compute the CDR (Cup-to-Disc) ratio that classify healthy or glaucoma infected fundus images of the eye.

Related work is described in Sect. 2. In Sect. 3, background concept of EfficientNet model is elaborated. In Sect. 4 demonstrations the suggested architecture-based DL model for glaucoma detection. Results description is based on benchmark datasets is illustrated in Sects. 5 and 6.

2 Related works

Some researchers implemented their system over the cloud to detect various diseases independent of location and experts for the benefit of society. Vilppu et al. [48], Ortega et al. [27], and Arnon Rosenthal et al. [32] proposed web-based applications for diagnosis of breast cancer, analysis of retinal fundus images and examination of body mass index (BMI). The cloud-based implementation of e-health application [3, 28] makes the system efficient, easy and economical too. Yin et al. [51] suggested a public cloud-based approach for automatic screening of medical images [42] based on the classification technique to detect glaucoma in the eye.

The CDR is an important factor for refereeing the presence of glaucoma in the eye. In past various methods has been suggested for automatic detection of optic cup and disc by segmentation of specific part of the eye. The optic disc region is brighter than its surroundings [40, 41]. The cloud architecture is proposed so that glaucoma detection system can be accessible without any geographical limitations.

The abnormalities due to glaucoma increases the optics disc size causing change in CDR. The optics nerves in the eye worked as carrier of visual signals from eye to brain. The healthy eye consists disc of 1.5 mm diameter and change in disc size reports abnormality [36]. The fundus image is segmented for separating optic disc using template-based method. Walter et al., used modified variance image for separating optic disc and rim from fundus images using watershed transform with 90% accuracy [49]. In this sequence, Chrástek et al. implemented robust scheme for separating the optic disc using nonlinear filtering, Hough transformation with accuracy 92% for localization and 82% for segmentation [11]. The efficiency in accurately segmenting the optic

disc area decides the efficiency of the proposed technique. Li et al. also suggested that Principal Component Analysis based algorithm for localization and segmentation with accuracy of 99% and 94% [21]. Lowell et al. [24] and Tobin et al. [47] uses template matching and probabilistic approach for the presence of glaucoma in the eye. Kande et al. implemented Bayesian classification for pixels of optic disc area and its modified version used points according to local variance to determine optic disc center and contour technique for boundary identification [20]. The state-of-the-art methods summaries various segmentation techniques for separating optic disc and cup area from the fundus image. Zhu et al. implement Sobel operator and Hugh transformation for segmentation of features in fundus image [53] with 90% efficacy on DRIVE dataset. Lu et al. reports 99.75% accuracy by segmenting boundaries of optic disc using circular transformation [25]. Later, other methods like vector based, wavelet-based techniques have been implemented for boundary detection of optics disc and reports 96.7% and 95.1% results accurately [1, 23].

DL is one the best tool for feature extraction and classification of image automatically. DL model learn by various test data images and based knowledge perceived; it classifies the images. Chen et al. implement the DL model on retinal fundus images for the same purpose to separate healthy and glaucoma infected eye images [8]. Later, Chen et al. [9] modifies the internal layer and used six layered DL followed by Abdullah et al. [2]. In 2017, Bharkad et al. improves the contrast of optic disc by overpowering the attention of blood vessels [7].

Optic-disc-to-cup-ratio is one of the prime attributes for glaucoma detection [37–39]. The manual inspection of eye is time expensive and need expertise too. Wong et al. utilizes bright pallor attribute for separation of optic sup [50]. Other methods have also been discussed for separating the optic cup from retinal fundus images via various techniques [5, 6, 21]. Juneja et al. recommended DL based automated system that segment that optic disc and cup to determine the healthy or glaucoma infected eye [19]. In this suggested model two Neural Network functions in collaboration to determine optic cup and disc. Gupta et al. uses the UNet for segmentation and efficient Net for classification purpose [17]. State of the art methods (SOTA) are described in Table 1.

3 Background concepts

In this suggested scheme, EfficientNet CNN and UNet++ is implemented for feature extraction and classification of glaucoma or healthy eye images. The architecture is implemented over cloud so that automatic system for the said purpose can be accessed independent of the geographical location.

CNN is fabricated with number of layers that can be customized depending on the requirement and size of the input. All layers are not necessarily implemented to all the networks. Number of layers improves the performance and also increases the computational time of the system. It learns and organized itself with or without supervision [8, 15, 16, 46]. Different SOTA methods have been implemented previously. The proposed paper uses EfficientNet [44] model as discussed in subsequent section.

Table 1 Result and discussion of various state of the art methods for glaucoma detection

SOTA methods	Methodology	Results	Dataset health eye/glaucoma infected eye
Chen et al. [10]	Sixed layered CNN	AUC* (83.1–88.7%)	SCES—46/1678 ORIGA—168/482
Al-Bander et al. [4]	23 Layered CNN and SVM	Acc.—88.20%, Spe. **—89.80% Sen. ***—85.00%,	RIM-ONE—200/255
Fu and Cheng [13]	4 -CNN with Ensemble Methodology	AUC* -91.83%, Spe. **—83.80% Sen. ***—84.78%	SCES—1636/46 ORIGA—168/482
Li et al. [22]	Pre-trained NN (Inception)	AUC* - 98.6%, Spe. **—92.00% Sen. ***—95.60%,	Personal—48116
Christopher et.al. [12]	VGG16, ResNet and Inception v3	AUC*—91%, Spe. **—88.00% Sen. ***—95.00%	Personal- 9189/5633
Shibata et.al. [34]	Pre-trained ResNet model	AUC*—96.5%	Personal—1768/1364
Soorya et al. [42]	Geometrical features, thresholding	Acc.—92.86%	Own dataset—364 Images
Mamta Juneja et al. [19]	Deep learning and CNN	Acc.—95.82%,	DRIVE-110
Abdel-Hamid [1]	Wavelet-based statistical and textural features	Glaucoma DB(Acc.—89.4 AUC*—92.2) HRF(Acc.—96.7 AUC*—94.7)	GlaucomaDB—66 High-resolution fundus (HRF)-30
Neeraj et al. [17]	CLAHE, EfficientNet, U-Net	Dice similarity (96%) IOU (91%)	DRISHTI-GS-31/70 RIM-ONE—455/311

Acc. Accuracy

*AUC (Area under Curve), **Spe. (Specificity), ***Sen. (Sensitivity)

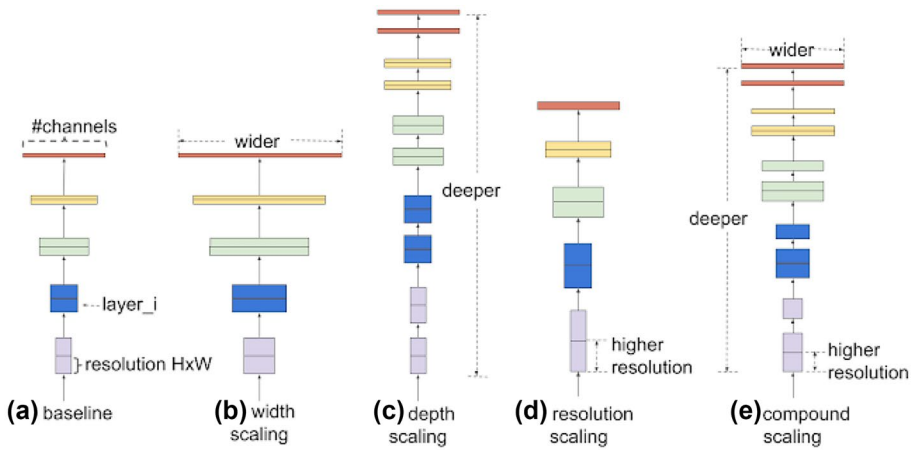


Fig. 3 Numerous scaling approaches assessment **a** baseline approach, **b–d** width-wise, depth-wise & resolution-wise scaling, **e** compound scaling method [44]

3.1 Architecture of EfficientNet

Google suggested afresh EfficientNet CNN having better accuracy and computational efficacy [44]. The detailed description of scaling is shown in Fig. 3. A new EfficientNet [44] model is used with accuracy and computational efficiency using framework like AutoML MNAS. EfficientNet architecture uses MBConv similar to MnasNet and MobileNetV2. Figure 4 represents the baseline network architecture of EfficientNet-B0.

3.2 Performance of EfficientNet

The efficiency of EfficientNet is compared on standard ImageNet [33] dataset with various CNN models and reports better accuracy and efficiency. EfficientNet CNN consists B0–B7 models. Table 2 represents various parameters needed for ImageNet dataset [33] with its accuracy. It is observed the tradeoff between size and accuracy of the model. The model scaling helps to improve the performance of the EfficientNet. It employs GoogleNet [43] or ResNet [18] models and modifies its design considering three techniques:

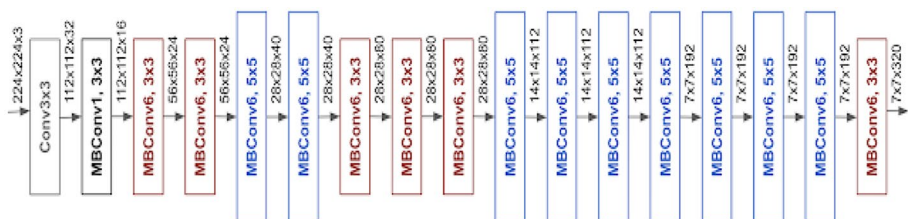


Fig. 4 EfficientNet-B0 baseline network [44]

Table 2 Description of EfficientNet-B0 [44]

Stage i	Operator K	Resolution $H_i \times W_i$	#Channels C_i	#Layer L_i
1	Conv 3×3	224×224	32	1
2	MBConv, $k3 \times 3$	112×112	16	1
3	MBConv, $k3 \times 3$	112×112	24	2
4	MBConv, $k5 \times 5$	56×56	40	2
5	MBConv, $k3 \times 3$	28×28	80	3
6	MBConv, $k5 \times 5$	14×14	112	3
7	MBConv, $k5 \times 5$	14×14	192	4
8	MBConv, $k3 \times 3$	7×7	320	1
9	Conv 1×1 & Pooling &	7×7	1280	1

- Alteration in the depth of the model.
- Alteration in the width of the model.
- Alteration in the resolution of an input image.

It is observed that above consideration scales up the accuracy for small models while its accuracy gets compact for larger models. Therefore, there must be tradeoff among depth, width and resolution for better performance.

3.3 Compound scaling of EfficientNet

Tan et al., [44] established tradeoff among depth, width and resolution for better performance known as compound scaling by using the coefficient ϕ as per mathematical equations below:

$$\text{Alternation in depth : } d = \alpha^\phi \quad (1)$$

$$\text{Alternation in width : } w = \beta^\phi \quad (2)$$

$$\text{Alternation in resolution : } r = \gamma^\phi \quad (3)$$

$$s.t. \alpha, \beta^2, \gamma^2 \approx 2 \quad (4)$$

$$\alpha \geq 1, \beta \geq 1, \gamma > 1 \quad (5)$$

α : depth β : width and γ resolution for the proposed model.

ϕ : Controlling coefficient utilizes in Eqs. 1–5.

3.4 EfficientNet-B0 model

The compound scaling using Eqs. 1–5 scaled up the CNN model. In Table 2, B0 the baseline EfficientNet model [44] was developed to improve the accuracy and FLOPS.

4 Proposed framework

Although various automated SOTA glaucoma detection techniques are giving quite good results in testing and validation of eye disease in screening the population. It is used at the local level by the standalone software and few trained people have access to that system in their ophthalmological clinic. To fulfil the need of the large population, the conventional system is not applicable. To reach to the masses, we propose a cloud-based real-time screening system for retinal fundus images for identifying the presence of glaucoma in the eye. The proposed method incorporates the concept of cloud Software as a Service (SaaS) and merges the features for the generation of medical reports.

4.1 Cloud-based setup for screening system

The objective of proposed cloud-based real-time glaucoma screening system is utilized for the automatic glaucoma screening to the masses. The proposed system is the blend of hybrid cloud screening system set up with the amalgamation of both the private and public cloud. The public cloud is used by the people to access the system for uploading the glaucoma suspect images. The private cloud effectively utilized to analyze the data and provide the screening assessment of glaucoma disease. In Fig. 5, the proposed framework is defined.

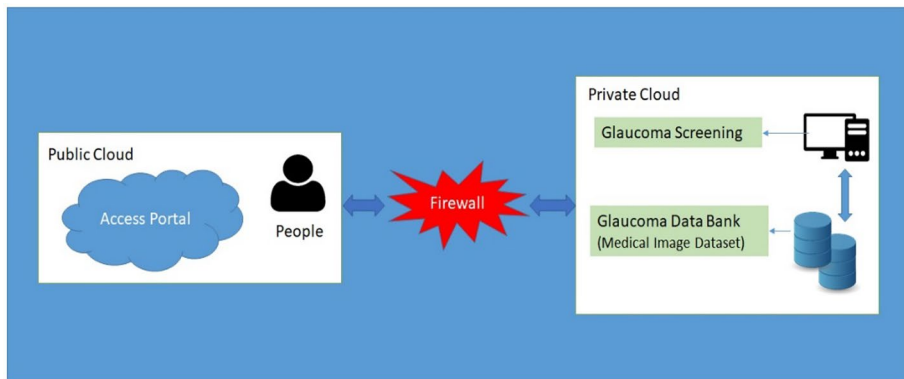
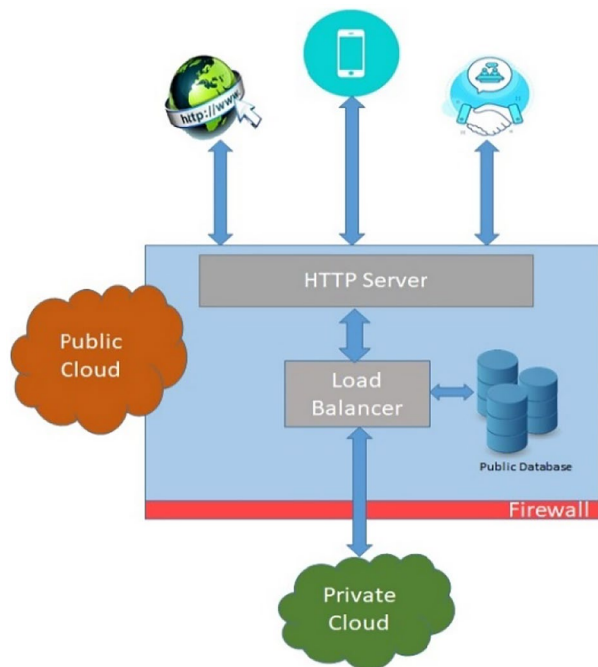


Fig. 5 A proposed cloud-based real-time framework for glaucoma screening system

Fig. 6 A public cloud framework

4.1.1 Public cloud framework: the access point

The retinal fundus images data have been collected from the different source are stored in the public cloud database. The public cloud is accessible to the people using three different ways: (1) Using the website, (2) Using mobile applications, (3) Using third-party tools. A public cloud framework is shown in Fig. 6. In this framework, the main two access points for common people are website and mobile applications which have fixed inputs and outputs. The customization task is available in the third-party tools.

4.1.2 Private cloud framework: the process point

The processing power is available in the private cloud as it is having high computing resources. The proposed algorithm for glaucoma detection using DL is available in a local cluster of private cloud. The information of glaucoma suspects and their analysis is available in the private cloud. The suspect can access their information from the personal account using the public cloud framework.

Algorithm 1: Encoder pipeline for either glaucoma infected eye or healthy eye

Step 1. Hyperparameters:

- i. $width = 512, height = 512$
- ii. $dropout = 0.2$ $input_shape = (height, width, 3)$

Step 2. Initializing the EfficientNet model B0 and load the pre-trained network, set its layers to trainable.

$h = \text{EfficientNetB0}(weights = 'imagenet', include_top = False)$

Step 3. Load the benchmark dataset with training, validation and test split for glaucoma infected and healthy eye.

Step 4. For loss, the Categorical_crossentropy is used.

Step 5. Model's lower convolutional layer's parameters (weights) are frozen.

$h.trainable = False$

Step 6. Now Model creation step. Add a global average pooling layer and a dense layer. Appended the classification subnetwork on the top of the baseline model.

- i. $model = model.sequential()$
- ii. $model.add(h)$
- iii. $model.add(layers.GlobalMaxPooling2D(name))$
- iv. $model.add(layers.Dense(2, activation = 'softmax', name = "fc_out"))$

Step 7. Model is now trained and compiled with different data augmentation choice through the training to decrease the chance of overfitting.

$model.compile(loss = 'categorical_crossentropy', optimizer, metrics)$

Step 8. To enhance the accuracy further, fine-tune hyperparameters and unfreeze a few layers of EfficientNet B0 Model.

Step 9. Finally, the deep learning model is trained for the binary classification either the Glaucoma infected Eye or Healthy Eye.

4.2 Glaucoma screening using deep learning

Though we have used the cloud-based system, the private cloud is responsible for the screening of the glaucoma suspect image uploaded by the people. The proposed framework is used to reveal glaucoma in retinal fundus images submitted by the people through the public cloud. The DL model is trained on the short number of samples in the datasets.

A large dataset such as ImageNet [33] has been utilized for the training of pre-trained network. There are 1000 classes labels in ImageNet [33]. The EfficientNet [44] has been used for classification of 1000 classes. The EfficientNet [44] is used as encoder for classification between glaucoma Infected Eye and Healthy Eye. In the proposed framework, pre-trained network is used with baseline EfficientNet-B0 model, which implement the technique of compound scaling, explained as below:

STEP 1: we initial fix $\varphi=1$, supposing twice more resources available, and do a trivial grid search of α , β , γ based on Eqs. 1–5. The best coefficient values for EfficientNet-B0 are $\alpha=1.2$, $\beta=1.1$, $\gamma=1.15$, underneath restriction of $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$.

Algorithm 2: Prediction of retinal fundus image for either glaucoma infected or healthy eye.

Step 1. Load the retinal fundus image I

Step 2. Preprocess the image I , apply masks, transforming it into fixed-size as $input_shape$ and add Gaussian noise and generate new retinal fundus image is I' .

$$I' = preprocess_input(I)$$

Step 3. The model trained in Algorithm 1 as Encoder and UNet++ as Decoder is used for the prediction and decode the classification based on the calculation of CDR.

$$y = model.predict(I')$$

Step 4. Now decode the prediction either healthy eye or glaucoma infected based on CDR. The value of CDR is greater than 0.5 than it is glaucoma infected eye otherwise healthy eye if CDR.

STEP 2: Finally, let α , β , γ as constants and scale-up baseline network B0 with different φ values using Eq. 3, to accomplish diverse models of EfficientNet-B1 to B7.

The image submitted by the people through the public cloud is submitted to the private cloud server for processing, the script is executed on the retinal fundus image. This comprises the completion of different image processing tasks. These involve the following segmentation: optic disc and optic cup. The goal is to compute the CDR which can be further utilized to categorize the glaucoma infected or healthy eye. The ratio of the areas of optic cup and disc can be utilized to compute the CDR which further results in existence of glaucoma. The value of CDR is greater than 0.5 than it is glaucoma infected eye otherwise healthy eye.

The UNet++ [52] has been used for semantic segmentation of optic disc and optic cup. In proposed framework the EfficientNet Model has been utilized as an encoder. According to width, resolution and depth EfficientNet is the balanced DL model. There are various versions of EfficientNet model it is ranging from B0 to B7 having parameters 5.3 million to 66 million respectively. In the proposed framework, U-Net++ is a decoder.

The five feature maps have been inserted from encoder to decoder. These feature maps have been utilized for the finding the segments. The algorithm 1 is used as an encoder pipeline by utilizing EfficientNet Model for glaucoma infected or Healthy eye in the private cloud based on CDR ratio computation. For doing the prediction of retinal fundus image submitted through the public cloud by the people is done through Algorithm 2, which is executed in the private cloud.

4.3 Implementation details of deep learning model

In this paper, we have used the baseline EfficientNet model [44] as encoder and UNet++ [52] as decoder for Segmentation. GlobalMaxPooling2D method is used to convert the tensor of size 4D to 2D, which moderately reduces the number of parameters of the network. In this paper we have used the Keras's Image Data Generator method for compiling

and training. Another way to make the representation of the model more significant using the fine-tuning concept. The generic and reusable features are present in earlier layers of the convolution base. However, special features are available in the higher layers. Implementation steps for the fine-tuning a network are as follows: firstly, append the encoder pipeline on the top of pre-trained baseline EfficientNet, baseline network should be freeze, train the classification subnetwork, further few layers are unfrozen in the baseline network. Later on, UNet++ is used as decoder pipeline. Finally, we have a fine-tuned model with improved validation accuracy. Table 3 demonstrates the summary of models having layers and parameters used in EfficientNet B0 and B1. For implementation, Python 3.7.6 has been utilized on the workstation having configuration Intel Xeon 5222 3.8 GHz with dual GPU NVIDIA™ Quadro RTX4000 having 8 GB GDDR5X RAM. The libraries which have been utilized are tensorflow, Keras, etc. The model was primarily trained for 90 epochs with batch size of 8. The learning rate for the model is 0.0001. On complete dataset, the loss was calculated after each training epoch. Further hyper parameters are tuned on the basis of loss generated in testing phase. The execution time of the proposed model for 90 epochs is 1678 s 451 ms.

4.4 Cloud-based glaucoma screening service framework

The proposed framework be responsible for providing automated services to the people. This is an integrated environment that unites people, screening services and Ophthalmology (eye doctor). Since the service is using the cloud platform its throughput is high. The proposed system has fewer human resources. Generally, this service framework is utilized for conducting large scale screening of glaucoma disease in the masses. The screening service framework based on cloud is depicted in Fig. 7.

In the framework, the people have to contact to the optical shops or fundus image collection Centre for getting their retinal fundus image. Once they collected submitted these retinal fundus images from secure login to the cloud platform through the website or mobile applications. The private cloud does the exhaustive screening of retinal fundus image using the DL algorithms available in the server. After that, the screening report has been generated. If the report is normal or showing healthy eye than information is directly provided to the people through their login. For the glaucoma infected eyes, the screening report has been sent to the Ophthalmology (eye doctor) for the expert advice. After that, the doctor advice has to be sent to the people. The whole process takes less time as compared to the manual methods.

5 Results analysis

In this paper, the DL algorithm of screening glaucoma in the private cloud has been evaluated on the standard datasets i.e., DRISHTI-GS [14, 40] and RIM-ONE [26]. Glaucoma infected and healthy images are available in these datasets.

5.1 Dataset

RIM-ONE [26]: A standard dataset for glaucoma diagnosis is designed by an ophthalmic image group. This is known as an Open Retinal Image Database. It is used for Optic Nerve Evaluation. For this dataset images are collected from three different Spanish

Table 3 Summary of models (a) EfficientNet B0, (b) EfficientNet B5 and (c) EfficientNet B7

Layer (type)	Output shape	Param #
(a) EfficientNet B0		
Efficientnet-b0 (model)		4049564
gap (GlobalMaxPooling2D)	(None, 5, 5, 1280)	0
drop_out (Dropout)	(None, 1280)	0
fc_out (Dense)	(None, 1280)	0
	(None, 2)	2562
Total parameter: 4,052,126 trainable parameters: 4,010,110 non-trainable parameters: 42,016		
(b) EfficientNet B5		
Efficient net-b5(model)		28510958
gap (GlobalMaxPooling2D)	(None, 5, 5, 1280)	0
drop_out (Dropout)	(None, 1280)	0
fc_out (Dense)	(None, 1280)	0
	(None, 2)	2562
Total Parameter: 28,513,520 trainable parameters: 28,340,784 non-trainable parameters: 1,72,736		
(c) EfficientNet B7		
Efficient net-b7 (model)		64095118
gap (GlobalMaxPooling2D)	(None, 5, 5, 1280)	0
drop_out (Dropout)	(None, 1280)	0
fc_out (Dense)	(None, 1280)	0
	(None, 2)	2562
Total Parameter: 64,097,680 trainable parameters: 63,786,960 non-trainable parameters: 310,720		

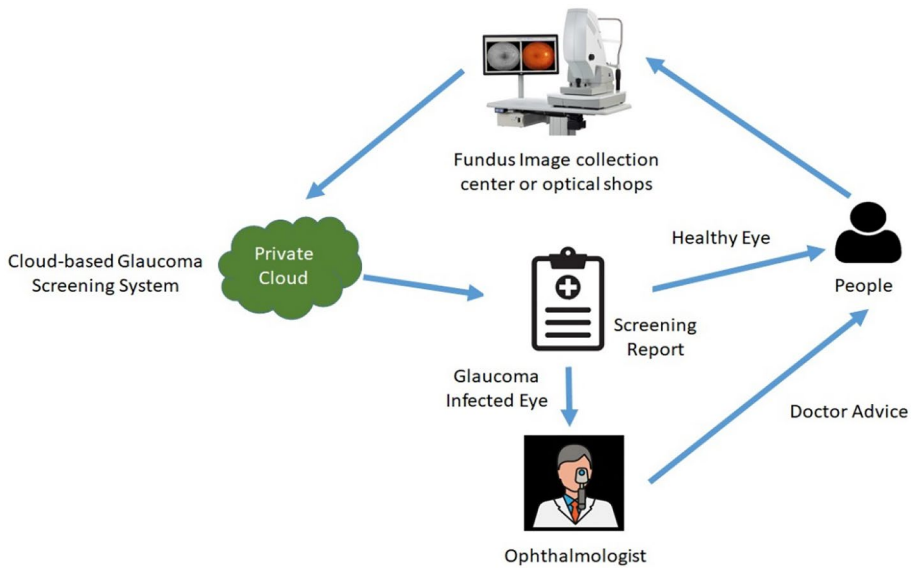


Fig. 7 The proposed cloud-based glaucoma screening service framework

hospitals and four glaucoma experts designed all images accordingly. A standard digital background camera Nidek AFC-210 is used to capture all images for this dataset. DRISHTI-GS [14, 40]: In this standard dataset, there are 101 retinal fundus images. These images are used for segmentation of optic disc. All images in this dataset are composed from dissimilar patients of Aravind Eye hospital in Madurai. The sample is collected from the age group of patients from 40 to 80 years. All the images are taken by a medical expert with dilated eyes. At the time of collection of images, the optic disk is set at the center with 30 degrees from view point. In this dataset, images are in PNG format. Table 4 defines the classification statistics on glaucoma infected vs. healthy eye.

5.2 Evaluation metrics

The proposed framework is quantified and compared with various baseline methods. For doing comparative analysis, various metrics have been used which are explained below in Table 5.

Table 4 Classification statistics on glaucoma infected vs. healthy eye

Name of dataset	Resolution of image	No. of images (glaucoma)	No. of images (healthy)	Image format
RIM-ONE r1	Not fixed	40	118	BMP
RIM-ONE r2	Not fixed	200	255	JPG
RIM-ONE r3	2144 × 1424	71	82	JPG
DRISHTI-GS	2996 × 1944	70	31	PNG

Table 5 Evaluation metrics used in proposed framework

S. No.	Evaluation metrics	Mathematical structure	Description
1	Mathews correlation coefficient (MCC)	$MCC = \frac{T.P \times T.N - F.P \times F.N}{\sqrt{(T.P + F.P)(T.P + F.N)(T.N + F.P)(T.N + F.N)}}$	T.P. aggregate true positives pixel, F.P aggregate false-positive pixel, T.N. aggregate true negative pixel, F.N. aggregate false-negative pixel X and Y are finite sets
2	Intersection over Union (IOU) or Jaccard index	$\text{Jaccard Index } (X, Y) = \frac{ X \cap Y }{ X \cup Y - X \cap Y } = \frac{ X \cap Y }{ max(X, Y) }$	No. of elements of sets A and B are A and B respectively
3	Dice coefficient for similarity	$\text{Dice score} = \frac{2 A \cap B }{ A + B }$	
4	Cohen kappa	$\text{Cohen Kappa} = \frac{2T.P + F.P + F.N}{\text{Tot.Acc} - \text{Rand.Acc.}}$ $\text{Tot. Acc} = \frac{T.P + T.N}{T.P + T.N + F.P + F.N}$ $\text{Rand. Acc.} = \frac{T.P + T.N}{(T.P + F.P)(T.N + F.N) + (T.P + F.P)(F.N + F.N)}$	T.P. aggregate true positives pixel, F.P aggregate false-positive pixel, T.N. aggregate true negative pixel, F.N. aggregate false-negative pixel
5	Pixel accuracy	$\text{Pix Acc} = \frac{T.P + T.N}{T.P + T.N + F.P + F.N}$	T.P. aggregate true positives T.N. aggregate true negative pixel
6	Structural similarity	$SS(a, b) = \frac{(2\mu_a \mu_b + c_1)(2\sigma_a \sigma_b + c_2)}{(\mu_a^2 + \mu_b^2 + c_1)(\sigma_a^2 + \sigma_b^2 + c_2)}$	μ_a is the average of a , μ_b is the average of b , σ_a^2 is the variance of a , σ_b^2 is the variance of b , σ_{ab} is the covariance of a and b $c1, c2$ are two variables used for stabilization

5.3 Comparison with the baseline methods

The quantifiable assessment of the performance of the algorithm for screening system available in the private cloud has been compared with the various SOTA techniques on the standard dataset. The evaluation metrics elaborated in Sect. 5.2 have been explored for the comparative study. For the segmentation of the optic cup and optic disc and finding their accuracy of the evaluation metrics are applied on the standard dataset. As depicted in Table 6 the accuracy of optic cup and disc. It is found that the DL algorithm of screening glaucoma in a private cloud using EfficientNet B7 outperforms the Juneja et al., as depicted in the bar chart shown in Fig. 8.

The DL algorithm of screening glaucoma in the private cloud is evaluated on the metric Dice similarity and IOU. As these evaluation metrics have been used by Neeraj et al. [17], Juneja et al. [19], Artem [53], Zilly et al., (a) [55] and Zilly et al., (b) [54] and their results are tabulated for Dice similarity and IOU are tabulated in Tables 6 and 7 respectively. It is visualized as bar chart in Fig. 9. The DL algorithm of screening glaucoma in the private cloud produces the EfficientNet B0, B5 and B7 as 0.92, 0.94 and 0.96 respectively for Dice similarity and the EfficientNet B0, B5 and B7 as 0.90, 0.91 and 0.93 respectively for IOU. It is found that the DL algorithm EfficientNet (B5 and B7) of screening glaucoma in the private cloud outperforms the baseline methods.

5.4 Cloud-based throughput analysis

The identification of glaucoma by the expert or ophthalmologist requires 5 to 10 min for analyzing the retinal fundus eye images. In general, these experts can analyze 50–100 glaucoma infected patient's eyes. In contrast, the proposed framework is very fast. It requires 20–30 s to process parallel 10 glaucoma infected eye images with the private cloud server having DL algorithms. So, the throughput of the private cloud server is approximately 30,000 images, which is very high.

5.5 Screening result report generation

The screening system generates the separate report of each submitted glaucoma suspect eye retinal fundus images. The report will be delivered to the individual through electronic media using cloud services on any available electronic device. These reports consist of the diagnosis of the individual with technical concern especially the CDR values and other diagnosis results. This report gives the individual to know the status of their eye as glaucoma suspect using this real time cloud-based approach.

5.6 Discussion about framework

In this paper, we have proposed a real time cloud-based framework for glaucoma screening using DL techniques. The framework is providing the automatic glaucoma screening of retinal fundus eye images. In this automatic screening, the analysis is done based on DL techniques available in private cloud and assessment report has been generated to send back to the individual or eye doctors. As this system is real time cloud-based the system delivers

Table 6 Different evaluation metrics of SOTA and proposed method for disc and cup accuracy

Evaluation metric	Accuracy of optic disc (Juneja, M et al.)	Accuracy of optic disc (Neeraj et al.)	Accuracy of optic cup (Neeraj et al.)	Accuracy of optic disc (proposed) Efficient-NetB0	Accuracy of optic cup (proposed) Efficient-NetB0	Accuracy of optic disc (proposed) Efficient-NetB5	Accuracy of optic cup (proposed) Efficient-NetB5	Accuracy of optic disc (proposed) Efficient-NetB7	Accuracy of optic cup (proposed) Efficient-NetB7
Matthews correlation coefficient (MCC)	0.9157	0.8454	0.9215	0.9187	0.8550	0.9208	0.8555	0.9210	0.8560
IOU	0.9062	0.8809	0.9213	0.9105	0.8908	0.9176	0.9010	0.9203	0.9036
Dice	0.9505	0.9361	0.9606	0.9566	0.9411	0.9588	0.9434	0.9597	0.9580
Cohen Kappa	0.9110	0.8340	0.9231	0.9219	0.8439	0.9231	0.8456	0.9231	0.8504
Pixel Accuracy	0.9586	0.9466	0.9654	0.9599	0.9501	0.9620	0.9663	0.9645	0.9671
Structural Similarity	0.8921	0.8690	0.9120	0.9019	0.8702	0.9056	0.8752	0.9101	0.8761

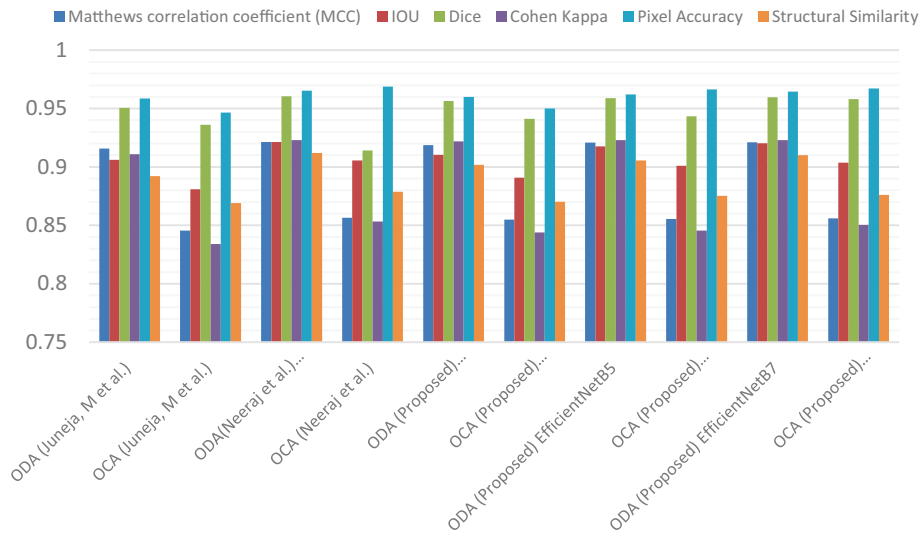


Fig. 8 The various evaluation metrics accuracy results of the optic disc and optic cup on standard datasets

Table 7 Dice similarity and IOU comparison of different SOTA and proposed methods

Methods	Dice similarity	IOU
Artem [53]	0.85	0.75
Zilly et al. (a) [55]	0.87	0.85
Zilly et al. (b) [54]	0.83	0.86
Juneja, M et al. [19]	0.93	0.88
Neeraj et al. [17]	0.96	0.91
Proposed (EfficientNet B0)	0.92	0.90
Proposed (EfficientNet B5)	0.94	0.91
Proposed (EfficientNet B7)	0.96	0.93

high accessibility to the common people using various means like website, mobile application etc. In the healthcare system, this framework advances the diagnosis which reaches to the masses and having the following features:

Anytime anywhere access: The screening system will be available to the common people because of the public cloud, which can be accessed anywhere and anytime. The screening report will also be delivered to the individual.

Accuracy and efficiency of the algorithm: The DL algorithm available in the private cloud for screening system has used lots of evaluation metrics. These metrics are Dice, IOU, F1-Score, Structural Similarity, Cohen Kappa, Pixel Accuracy and MCC are explained in Sect. 5.2, they provide the cloud-based system with the power of screening the retinal fundus images. The results of these metrics for the proposed model are elaborated in Sect. 5.3.

User interfaces with cross-platform support: The easy, interactive and user-friendly interface is given to the people so that they can interact with the glaucoma screening system.

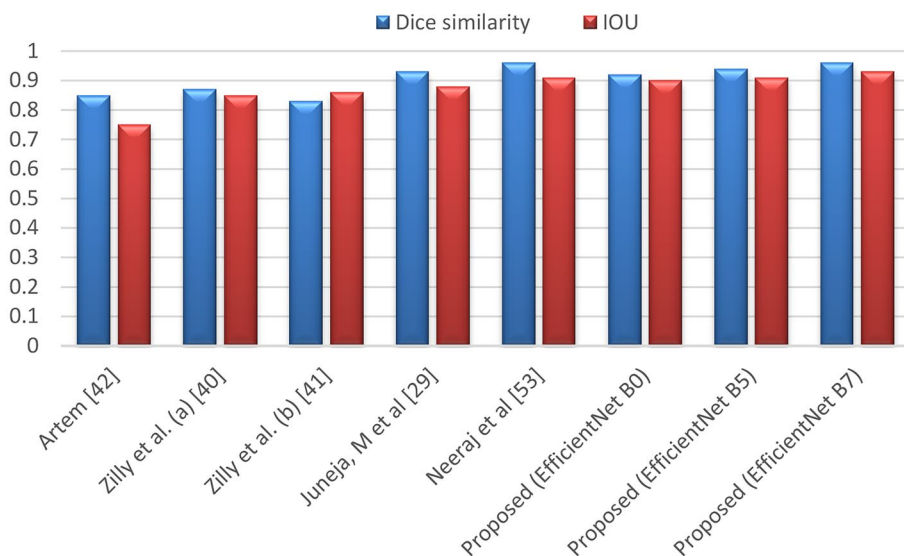


Fig. 9 The experimental result with Dice similarity and IOU for cup segmentation on DRISHTI-GS dataset

The screening system will be accessed on different electronic devices having browser like Firefox, Chrome and Internet Explorer etc. It is also accessible to the mobile devices having Android or iOS platform.

6 Conclusion

Every year the glaucoma disease affected lot of people worldwide. It causes permanent blindness if not detected at early stage. The traditional technique for its diagnosis is physically done by the eye specialist or ophthalmologists. This traditional and conventional screening system requires a lot of trained human resources. For developing countries, medical resources are very limited. Hence some healthcare system which is real-time, automatic, efficient, fast, and effective framework is to be developed for glaucoma screening system. In this paper, a real time cloud-based framework is proposed for glaucoma screening using a DL technique for the classifying the glaucoma infected image using EfficientNet and UNet++. The people can access it using the public cloud. The algorithm is available in the private cloud which is trained on the benchmark dataset DRISHTI-GS, RIM-ONE. The investigational results demonstrate that the proposed framework outperformed the other SOTA models for glaucoma detection. In this automatic screening, the analysis is done based on DL techniques available in private cloud and assessment report has been generated to send back to the individual or eye doctors. Henceforth, this framework advances the diagnosis which reaches to the masses.

References

1. Abdel-Hamid L (2020) Glaucoma detection from retinal images using statistical and textural wavelet features. *J Digit Imaging* 33(1):151–158

2. Abdullah M, Fraz MM, Barman SA (2016) Localization and segmentation of optic disc in retinal images using circular Hough transform and grow-cut algorithm. *PeerJ* 4:e2003
3. Aguilar-Rivera M, Erudaitius DT, Wu VM, Tantiogloc JC, Kang DY, Coleman TP et al (2020) Smart electronic eyedrop bottle for unobtrusive monitoring of glaucoma medication adherence. *Sensors* 20(9):2570
4. Al-Bander B, Al-Nuaimy W, Al-Tae MA, Zheng Y (2017) Automated glaucoma diagnosis using deep learning approach. In: 2017 14th International multi-conference on systems, signals devices (SSD). IEEE, Marrakech, pp 207–210
5. Almazroa A, Alodhayb S, Raahemifar K, Lakshminarayanan V (2017) Optic cup segmentation: type-II fuzzy thresholding approach and blood vessel extraction. *Clin Ophthalmol* (Auckland, NZ) 11:841
6. Arnay R, Fumero F, Sigut J (2017) Ant colony optimization-based method for optic cup segmentation in retinal images. *Appl Soft Comput* 52:409–417
7. Bharkad S (2017) Automatic segmentation of optic disk in retinal images. *Biomed Signal Process Control* 31:483–498
8. Chen X, Xu Y, Yan S, Wong DW, Wong TY, Liu J (2015) Automatic feature learning for glaucoma detection based on deep learning. In: International conference on medical image computing and computer-assisted intervention 2015 Oct 5. Springer, Cham, pp 669–677
9. Chen X, Xu Y, Wong DW, Wong TY, Liu J (2015) Glaucoma detection based on deep convolutional neural network. In: Engineering in medicine and biology society (EMBC), 2015 37th annual international conference of the IEEE 2015 Aug 25. IEEE, Milan, pp 715–718
10. Chen X, Xu Y, Wong DWK, Wong TY, Liu J (2015) Glaucoma detection based on deep convolutional neural network. In: 2015 37th Annual international conference of the IEEE engineering in medicine and biology society (EMBC). IEEE, Milan, pp 715–718
11. Chrastek R, Niemann H, Kubecka L, Jan J, Derhartunian V, Michelson G (2005) Optic nerve head segmentation in multimodal retinal images. In: Medical imaging 2005: image processing, vol 5747. International Society for Optics and Photonics, Bellingham, pp 1604–1615
12. Christopher M, Belghith A, Bowd C, Proudfoot JA, Goldbaum MH, Weinreb RN, Girkin CA, Liebmann JM, Zangwill LM (2018) Performance of deep learning architectures and transfer learning for detecting glaucomatous optic neuropathy in fundus photographs. *Sci Rep* 8(1):16685
13. Fu H, Cheng J, Xu Y, Zhang C, Wong DWK, Liu J, Cao X (2018) Disc-aware ensemble network for glaucoma screening from fundus image. *IEEE Trans Med Imaging* 37(11):2493–2501
14. Fumero F, Alayón S, Sanchez JL, Sigut J, Gonzalez-Hernandez M (2011) RIM-ONE: an open retinal image database for optic nerve evaluation. In: 2011 24th international symposium on computer-based medical systems (CBMS). IEEE, Bristol, pp 1–6
15. Gupta N, Jalal AS (2019) Integration of textual cues for fine-grained image captioning using deep CNN and LSTM. *Neural Comput Appl* 32:17899
16. Gupta N, Jalal AS (2019) A robust model for salient text detection in natural scene images using MSER feature detector and Grabcut. *Multimed Tools Appl* 78(8):10821–10835
17. Gupta N, Garg H, Agarwal R (2021) A robust framework for glaucoma detection using CLAHE and EfficientNet. *Vis Comput*. <https://doi.org/10.1007/s00371-021-02114-5>
18. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition. IEEE, Piscataway, pp 770–778
19. Juneja M, Singh S, Agarwal N, Bali S, Gupta S, Thakur N, Jindal P (2019) Automated detection of glaucoma using deep learning convolution network (G-net). *Multimed Tools Appl* 79:15531
20. Kande GB, Subbaiah PV, Savithri TS (2008) Segmentation of exudates and optic disk in retinal images. In: Computer vision, graphics & image processing. ICVGIP'08. Sixth Indian conference on 2008 Dec 16. IEEE, Bhubaneswar, pp 535–542
21. Li H, Chutatape O (2003) A model-based approach for automated feature extraction in fundus images. In: Null. IEEE, Nice
22. Li Z, He Y, Keel S, Meng W, Chang RT, He M (2018) Efficacy of a deep learning system for detecting glaucomatous optic neuropathy based on color fundus photographs. *Ophthalmology* 125(8):1199–1206
23. Liu SP, Chen J (2011) Detection of the optic disc on retinal fluorescein angiograms. *J Med Biol Eng* 31(6):405–412
24. Lowell J, Hunter A, Steel D, Basu A, Ryder R, Fletcher E, Kennedy L (2004) Optic nerve head segmentation. *IEEE Trans Med Imaging* 23(2):256–264
25. Lu S (2011) Accurate and efficient optic disc detection and segmentation by a circular transformation. *IEEE Trans Med Imag* 30(12):2126–2133
26. Mehdizadeh M, Dolatyar S (2009) Study of effect of adaptive histogram equalization on image quality in digital preapical image in pre apex area. *J Biol Sci* 4(8):922–924

27. Ortega M, Barreira N, Novo J, Penedo MG, Pose-Reino A, Gomez-Ulla F (2010) Sirius: a web-based system for retinal image analysis. *Int J Med Inform* 79(10):722–732
28. Owoyemi A, Owoyemi J, Osiyemi A, Boyd A (2020) Artificial intelligence for healthcare in Africa. *Front Digit Health* 2:6
29. Palaniappan K, Bunyak F, Chaurasia SS (2019) Image analysis for ophthalmology: segmentation and quantification of retinal vascular systems. In: Guidoboni G, Harris A, Sacco R (eds) *Ocular fluid dynamics. Modeling and simulation in science, engineering and technology*. Birkhäuser, Cham
30. Pallawala P, Hsu W, Lee ML, Eong KGA (2004) Automated optic disc localization and contour detection using ellipse fitting and wavelet transform. *ECCV*. Springer, Berlin, Heidelberg, pp 139–151
31. Quigley HA, Broman AT (2006) The number of people with glaucoma worldwide in 2010 and 2020. *Br J Ophthalmol* 90(3):262–267
32. Rosenthal A, Mork P, Li MH, Stanford J, Koester D, Reynolds P (2010) Cloud computing: a new business paradigm for biomedical information sharing. *J Biomed Inform* 43(2):342–353
33. Russakovsky O, Deng J, Su H, Krause J, Satheesh S, Ma S et al (2015) Imagenet large scale visual recognition challenge. *Int J Comput Vision* 115(3):211–252
34. Shibata N, Tanito M, Mitsuhashi K, Fujino Y, Matsuura M, Murata H, Asaoka R (2018) Development of a deep residual learning algorithm to screen for glaucoma from fundus photography. *Sci Rep* 8(1):14665
35. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*
36. Singh LK, Garg H (2019) Detection of glaucoma in retinal fundus images using fast fuzzy C means clustering approach. In: 2019 International conference on computing, communication, and intelligent systems (ICCCIS). IEEE, Greater Noida, pp 397–403
37. Singh LK, Garg H (2020) Automated glaucoma type identification using machine learning or deep learning techniques. In: *Advancement of machine intelligence in interactive medical image analysis*. Springer, Singapore, pp 241–263
38. Singh LK, Garg H, Khanna M, Bhadoria RS (2021) An enhanced deep image model for glaucoma diagnosis using feature-based detection in retinal fundus. *Med Biol Eng Comput* 59(2):333–353
39. Singh LK, Garg H, Khanna M (2021) An artificial intelligence-based smart system for early glaucoma recognition using OCT images. *Int J E-Health Med Commun (IJEHMC)* 12(4):32–59
40. Sivaswamy J, Krishnadas SR, Joshi GD, Jain M, Tabish AU (2014) Drishti-gs: retinal image dataset for optic nerve head (ONH) segmentation. In: 2014 IEEE 11th international symposium on biomedical imaging (ISBI). IEEE, Beijing, pp 53–56
41. Sivaswamy J, Krishnadas S, Chakravarty A, Joshi GD, Tabish AS (2015) A comprehensive retinal image dataset for the assessment of glaucoma from the optic nerve head analysis. *JSM Biomed Imaging Data Papers* 2(1):1004
42. Soorya M, Issac A, Dutta MK (2019) Automated framework for screening of glaucoma through cloud computing. *J Med Syst* 43(5):136
43. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D et al (2015) Going deeper with convolutions. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. IEEE, Piscataway, pp 1–9
44. Tan M, Le QV (2019) Efficientnet: rethinking model scaling for convolutional neural networks. *arXiv preprint arXiv:1905.11946*
45. Tessier-Lavigne M (2000) Visual processing by the retina. *Principles of neural science*. McGraw-Hill, New York, pp 507–522
46. Thakur N, Juneja M (2017) Clustering based approach for segmentation of optic cup and optic disc for detection of glaucoma. *Curr Med Imaging Rev* 13(1):99–105
47. Tobin KW Jr., Chaum E, Govindasamy VP, Karnowski TP, Sezer O (2006) Characterization of the optic disc in retinal imagery using a probabilistic approach. In: *Medical imaging 2006: image processing*, vol 6144. International Society for Optics and Photonics, Bellingham
48. Tuominen VJ, Ruotoistenmäki S, Viitanen A, Juppunen M, Isola J (2010) ImmunoRatio: a publicly available web application for quantitative image analysis of estrogen receptor (ER), progesterone receptor (PR), and Ki-67. *Breast Cancer Res* 12(4):R56
49. Walter T, Klein JC, Massin P, Erginay A (2002) A contribution of image processing to the diagnosis of diabetic retinopathy-detection of exudates in color fundus images of the human retina. *IEEE Trans Med* 21:1236–1243
50. Wong DWK, Liu J, Lim JH, Jia X, Yin F, Li H, Wong TY (2008) Level-set based automatic cup-to-disc ratio determination using retinal fundus images in ARGALI. In: 2008 30th Annual international conference of the IEEE engineering in medicine and biology society. IEEE, Vancouver, pp 2266–2269

51. Yin F, Wong DWK, Quan Y, Yow AP, Tan NM, Gopalakrishnan K et al (2015) A cloud-based system for automatic glaucoma screening. In: 2015 37th Annual international conference of the IEEE engineering in medicine and biology society (EMBC). IEEE, Milan, pp 1596–1599
52. Zhou Z, Siddiquee MMR, Tajbakhsh N, Liang J (2018) UNet++: a nested U-Net architecture for medical image segmentation. In: Deep learning in medical image analysis and multimodal learning for clinical decision support. Springer, Cham, pp 3–11
53. Zhu X, Rangayyan RM, Ells AL (2010) Detection of the optic nerve head in fundus images of the retina using the Hough transform for circles. *J Digit Imaging* 23(3):332–341
54. Zilly JG, Buhmann JM, Mahapatra D (2015) Boosting convolutional filters with entropy sampling for optic cup and disc image segmentation from fundus images. In: International workshop on machine learning in medical imaging. Springer, Cham, pp 136–143
55. Zilly J, Buhmann JM, Mahapatra D (2017) Glaucoma detection using entropy sampling and ensemble learning for automatic optic cup and disc segmentation. *Comput Med Imaging Graph* 55:28–41

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